Chemosphere 183 (2017) 97–106

Contents lists available at ScienceDirect

Chemosphere

journal homepage: www.elsevier.com/locate/chemosphere

Review

Do polychlorinated biphenyls cause cancer? A systematic review and meta-analysis of epidemiological studies on risk of cutaneous melanoma and non-Hodgkin lymphoma

Claudia Zani*, Elisabetta Ceretti, Loredana Covolo, Francesco Donato

Department of Medical and Surgical Specialities, Radiological Sciences and Public Health, University of Brescia, Viale Europa 11, 25123, Brescia, Italy

HIGHLIGHTS

- In 2015 IARC upgraded the classification of PCBs to carcinogenic to human.
- The studies on PCB exposure and risk of cancer provided discrepant evidences.
- Our results don't support the hypothesis that PCBs cause melanoma and/or NHL.

ARTICLE INFO

Article history:
Received 18 January 2017
Received in revised form
20 April 2017
Accepted 9 May 2017
Available online 12 May 2017

Handling Editor: A. Gies

Keywords:
Polychlorinated biphenyls
PCB
Cancer
Melanoma
Non-Hodgkin lymphoma
Review
Meta-analysis

ABSTRACT

In 2015 a IARC Working Group upgraded the classification of PCBs to Group 1 “Carcinogenic to humans”, also on the basis of evidence from epidemiological studies showing an excess risk for melanoma. Increased risks for non-Hodgkin lymphoma (NHL) and breast cancer were also reported though the evidence was limited. However, some recent reviews of studies on PCB exposure and risk of cancer provided discrepant findings. Therefore, we re-evaluated the association between exposure to PCBs and risk of melanoma and NHL by a systematic review and meta-analysis. We retrieved 11 independent cohort studies on occupationally exposed workers. About half of them showed increased standardized mortality or incidence ratios (SMRs or SIRs) for melanoma and none for NHL. The pooled SMRs were 1.32 (95% CI: 1.05–1.64) for melanoma and 0.94 (0.73–1.23) for NHL. Among population-based cohort and case-control studies with individual measures of PCB exposure, one only study was carried out on PCB exposure and melanoma, showing an odds ratio (OR) of 6.0 (2.0–18.2) for the highest compared to lowest quartile of PCB distribution. 13 cohort and case-control studies evaluated the association between NHL and PCB concentration in blood or subcutaneous fat, with summary OR = 1.5 (1.1–1.7) for the highest vs lowest quantile of PCB distribution. However, two cohort studies on people intoxicated by rice oil containing PCBs found no excess of deaths for skin cancer and inconsistent results for NHL. In conclusion, these findings do not provide a strong evidence that PCB exposure can increase the risk of melanoma and NHL in humans.

© 2017 Elsevier Ltd. All rights reserved.

Contents

1. Introduction 98
2. Methods 98
3. Results 99
   3.1. Occupational cohort studies on exposure to PCBs and cutaneous melanoma 99
   3.2. Population-based studies on exposure to PCBs and melanoma with individual measures of exposure 99
   3.3. Occupational cohort studies on exposure to PCBs and non-Hodgkin lymphoma 101
   3.4. Population-based studies on exposure to PCBs and non-Hodgkin lymphoma 101

* Corresponding author.
E-mail address: claudia.zani@unibs.it (C. Zani).

http://dx.doi.org/10.1016/j.chemosphere.2017.05.053
0045-6535/© 2017 Elsevier Ltd. All rights reserved.

4. Discussion . . . 102
4.1. Cutaneous melanoma . . . 102
4.2. Non-Hodgkin lymphoma . . . 103
4.3. A comparison of this review with the IARC Working Group evaluation and the Boffetta et al. meta-analysis . . . 104
Disclosure statement . . . 105
Acknowledgements . . . 105
Supplementary data . . . 105
References . . . 105

1. Introduction

Polychlorinated biphenyls (PCBs) are organic compounds containing from one to ten chlorine atoms attached to a biphenyl nucleus, including 209 different congeners, largely produced as dielectric fluids in capacitors, transformers and other materials since the 1930s to the 1980s (ATSDR, 2003). After that date, their negative impact on the environment was apparent and subsequently their production and use were banned in most countries. Due to their long persistence in the environment and their bioaccumulation along the food chain, they have diffused everywhere and have contaminated virtually all human populations. Their possible effects on human health are a matter of concern and a large bulk of epidemiologic and toxicological research is available at present.

In 2016, the International Agency for Research on Cancer (IARC) upgraded the classification of the PCBs to Group 1 "Carcinogenic to humans" from the previous Group 2A classification "Probably carcinogenic to humans" (IARC, 2016). These compounds were classified in Group 1 on the basis of sufficient evidence of carcinogenicity in humans and experimental animals, and of the evidence of an aryl hydrocarbon receptor (AhR)-mediated mechanism of carcinogenicity for some PCB congeners similar to 2,3,7,8-tetrachlorodibenzopara-dioxin (dioxin-like PCBs) in both humans and experimental animals (Lauby-Secretan et al., 2013; IARC, 2016). The IARC Working Group concluded for sufficient evidence of PCB carcinogenicity for melanoma and limited evidence for non-Hodgkin lymphoma (NHL) and breast cancer. However, a recent meta-analysis found no evidence of association between PCB exposure and the risk of malignant melanoma (Boffetta et al., 2016). Two reviews on PCB exposure and the risk of NHL found contrasting results (Golden and Kimbrough, 2009; Freeman and Kohles, 2012). Another review of epidemiological studies on PCB exposure and risk of cancer showed some evidence for a possible role of PCB exposure in the development of NHL but inconsistent results for other cancers (Zani et al., 2013).

Because of these discrepancies between the IARC evaluation and the results of some reviews, we aimed to re-evaluate the association between exposure to PCBs and risk of cutaneous melanoma and NHL by a systematic review and meta-analysis.

2. Methods

We updated our previous search (Zani et al., 2013) considering also the studies cited in the IARC Monograph. The methods of this systematic review matched those described in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

The literature search covered the period January 2013–December 2016 and it was carried out using electronic databases of scientific research (PubMed, Scopus, Web of Science).

We conducted the bibliographic search using the following query strings for each electronic database: PubMed: *(((PCB[All Fields] OR ("polychlorinated biphenyls"[MeSH Terms] OR ("polychlorinated"[All Fields] AND "biphenyls"[All Fields]) OR "polychlorinated biphenyls"[All Fields])) OR organohalogen[All Fields]) AND ((("tumours"[All Fields] OR "neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "tumors"[All Fields]) OR ("neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "cancer"[All Fields])) OR ("neoplasms"[MeSH Terms] OR "neoplasms"[All Fields] OR "neoplasm"[All Fields]))) AND (((cohort[All Fields] OR case-control[All Fields]) OR follow-up[All Fields]) OR longitudinal[All Fields]) AND ("2013/01/01"[PDAT]: "3000"[PDAT]) AND english[Language]))*

Web of Science: *(TOPIC:(((((PCB OR polychlorinated biphenyls) OR organohalogen) OR organochlorine) AND ((tumours OR cancer) OR neoplasm)) AND (((cohort OR case-control) OR follow-up) OR longitudinal)) AND LANGUAGE:(English))*

Scopus: *((PCB OR polychlorinated biphenyls OR organohalogen OR organochlorine) AND (tumour OR cancer OR neoplasm) AND (cohort OR case-control OR follow-up OR longitudinal))*

We selected articles according to the following criteria: studies performed in humans, written in English, full text available, key terms present anywhere in the paper, articles with original data. We considered only a) occupational cohort studies and b) population-based cohort or case-control studies with direct measures of exposure to PCBs (in biologic samples) that allowed the computation of standardized mortality or morbidity ratios (SMR), relative risks (RR), odds ratios (OR) or other measures of association. Therefore we excluded studies with only indirect measures of exposure, such as place of residence, occupational history or PCB measurements in environmental matrices, animals or plants.

Studies investigating various chemicals, including PCBs, dioxins or other organochlorine, were considered only if separated measures for PCBs were available. We considered only studies that showed data on melanoma and Non Hodgkin Lymphoma.

We also included two cohort studies of Asian people undergoing mass poisoning by PCBs because both measures of total intake and of serum concentration of these chemicals were available, though not for each individual. Therefore, these cohorts were considered separately from the cohort and case-control studies with individual measures of PCBs.

Related articles were also searched for, so as the cited papers in the articles retrieved and in reviews. Two authors (C.Z. and F.D.) examined separately titles, abstracts and each paper included.

To avoid duplicate data, we considered only the most recent update or the most complete of two or more studies performed on the same population.

Among occupational cohort studies, only those reporting standardized mortality ratio (SMR), proportional mortality ratio (PMR) or standardized incidence ratio (SIR) were used for the meta-analysis, with a summary SMR based on the number of observed and expected deaths or cases in each study, as suggested by Wong and Raabe (1996). Studies reporting no case or death could not be used for calculating the combined SMRs. Studies that provided data only for all lymphopoietic malignancies together, including leukemias, were excluded, whereas those reporting lymphomas (NHL

and Hodgkin Lymphoma) or lymphoid malignancies (NHL and Hodgkin Lymphoma and multiple myeloma) together were used, because NHL are usually the majority of total lymphomas. Similarly, we used mortality data for "skin cancer" when melanoma was not reported separately, because this is the most common cause of death among all skin cancers. Only data on the whole cohorts were included in the meta-analysis. If only separate data for men and women were reported, we computed SMRs, with their 95% confidence intervals (95% CIs), for total subjects, using the tabulated values of 95% confidence limits limit factors for estimating a Poisson-distributed variable (Breslow and Day, 1987). Subgroup analyses according to duration of employment, lag-time or cumulative exposure were not considered. When more occupational cohorts were put together, we considered separately the cohort of workers at each plant as the study unit.

Log-transformed 95% CIs were calculated for estimating the standard errors (SEs) of the ln (SMR) by the formula: SE of ln (SMR) = [ln (upper limit) − ln (lower limit)]/2 × 1.96) (Bradburn, 2004). Summary SMRs and 95% CIs were calculated using a random-effects model (Dersimonian and Laird, 1986). Heterogeneity of effects across studies was assessed by the Cochran's Q statistic and was deemed significant when p value was <0.10. To evaluate the percentage of variation attributable to heterogeneity, we calculated the I^2 statistics (Higgins and Thompson, 2002). Substantial heterogeneity was defined at $I^2 > 50\%$ and/or p value < 0.10 by the Q test.

Publication bias was evaluated using the funnel plot method displaying SMRs from individual studies versus their precision (1/standard error) and confirmed using Egger's test method (Egger et al., 1998). Sensitivity analysis was carried out to evaluate the influence of individual studies on the summary effect estimate (Tobias, 1999).

A p value < 0.05 at two sided tests was considered statistically significant. Statistical analyses were conducted using STATA 12.0 (Stata Corp, College Station, TX), notably the metan commands.

3. Results

The present analysis is based on an update of our previous research (Zani et al., 2013). Following the Prisma flow diagram (reported in Supplementary material), we retrieved a total of 184 studies from January 2013 to December 2016: 90 from Web of Science, 48 from PubMed and 46 from Scopus. After eliminating duplicates, we included 121 studies. A total of 21 full text articles were assessed for eligibility: nine of them reported data on cancer, but only one contained data about NHL and melanoma (Pesatori et al., 2013). Another article on PCB exposure and melanoma was retrieved during the revision of the present paper and included in the review (Kelly et al., 2017).

3.1. Occupational cohort studies on exposure to PCBs and cutaneous melanoma

We retrieved 27 cohort studies on workers in capacitor and transformer producing factories, in electrical power, and in transformer manufacture and telecommunications industry. After eliminating duplicates and separating cohorts for each plant, we included 11 studies (De Guire et al., 1992; Tynes et al., 1994; Gustavsson and Hogstedt, 1997; Loomis et al., 1997; Robinson et al., 1999; Yassi et al., 2003; Mallin et al., 2004; Pesatori et al., 2013; Ruder et al., 2014; three cohorts).

In Table 1, we reported the characteristics of these studies, the SMRs or SIRs and their 95% CIs for all cancers and cutaneous melanoma. Almost all studies used mortality as a measure of effect, and the general population of the area or the whole country for computing the expected events. All measures were age-standardized, and separate analyses for men and women were performed in some studies. Most studies included workers who had worked at the factory for at least 90 days. The Illinois cohort was also exposed to other chemicals, some of which are potential carcinogens, such as chlorinated naphthalene and phenoxy herbicides (Mallin et al., 2004). For the three US National Institute for Occupational Safety and Health (NIOSH) cohorts of capacitor manufacturing workers (Indiana, Massachusetts and New York), separate data for each plant were used, as described in Methods, which were referred to workers with at least 3 months of working in the factory, as reported in Ruder et al. (2014).

Only two studies reported a statistically significant excess of deaths or cases for all cancer (Robinson et al., 1999; Mallin et al., 2004). As regards cutaneous melanoma, four studies showed a significantly increased risk (Robinson et al., 1999; Yassi et al., 2003; Ruder et al., 2014 in Indiana and New York cohorts). Two studies found an increased risk of melanoma of the skin for some subgroups of workers only: those with lag-time > 20 years and >10000 h of cumulative exposure in the study by Loomis et al. (1997) (RR = 4.8; 95% CI: 1.5–15.1); men, though not women, with lag-time < 20 years in the study by De Guire et al. (1992) (SMR = 9.4; 95% CI: 1.1–34.1). The remaining five studies showed no excess of deaths or cases of skin melanoma compared to those expected (Tynes et al., 1994; Gustavsson and Hogstedt, 1997; Mallin et al., 2004; Pesatori et al., 2013; Ruder et al., 2014; Massachusetts cohort).

Data on risk of melanoma by duration of working in the factory was reported in few studies: two of them showed an increased risk with increasing duration (Tynes et al., 1994; Loomis et al., 1997), whereas the joined analysis of the three NIOSH cohorts did not show a trend of risk for melanoma with increasing cumulative exposure, with a relative risk of 0.20 (95% CI: 0.06–0.70) for the highest level of estimated exposure (Ruder et al., 2014).

The meta-analysis of risk of skin melanoma in occupational cohorts is shown in Fig. 1. All the studies set out in Table 1 were included except one (Gustavsson and Hogstedt, 1997) reporting no case of melanoma. The pooled SMR, based on a random-effects model, was 1.32 (95% CI: 1.05–1.64; p = 0.03) with a moderate evidence of heterogeneity across studies ($I^2 = 52.5\%$, p = 0.03). The funnel plot displaying the logarithm of the SMR versus its standard error (figure not shown) and the Egger's test (p = 0.4) showed no evidence of publication bias. The sensitivity analysis showed that no single study influenced the overall SMR substantially (data not shown). The inclusion of the pooled data of the three NIOSH cohorts together rather than separately, as reported by Ruder et al. (2014) did not change the results significantly (pooled SMR = 1.17, 95% CI: 1.04–1.30; p = 0.008).

3.2. Population-based studies on exposure to PCBs and melanoma with individual measures of exposure

Only one case-control study was retrieved on melanoma and PCB serum levels (Gallagher et al., 2011) (Table 2). A total of 14 PCB congeners were determined and total PCBs were computed as the sum of single congeners in 80 cases and 310 controls. An increased risk of melanoma for the highest compared to the lowest quartile of PCB distribution was found (OR: 6.0, 95% CI: 2.0–18.2) after controlling for the major risk factors for this disease, i.e. sun sensitivity and sun exposure. The association was stronger for total non-dioxin-like PCB levels (OR = 7.0; 95% CI: 2.3–21.4) than for total dioxin-like PCB levels (OR: 2.8, 95% CI: 1.0–8.0) comparing the highest versus lowest quartile.

A high exposure to PCBs, and polychlorinated dibenzofurans (PCDFs) occurred in people who ingested rice oil contaminated by

Table 1
Occupational cohorts among workers exposed to polychlorinated biphenyls: mortality for, or incidence of, all cancers, skin melanoma and non-Hodgkin lymphoma (NHL).

Authors, year	Location of the plant	Factory	No. subjects	Period of follow-up	Cancer (No. deaths or cases): SMR (95% CI)
De Guire et al., 1992	Canada	Telecommunications industry	9590	1976–1983	All cancers (n = 84): 0.6 (0.5–0.8)* Melanoma (n = 3): 2.5 (0.5–7.2)* Lymphatic and haematopoietic (males, n = 7): 0.7 (0.3–1.5)
Tynes et al., 1994	Norway	Electrical power	5088 (M)	1953–1985	All cancers (incidence: n = 486): SIR = 1.0 (0.9–1.1) Melanoma (incidence: n = 19): SIR = 1.1 (0.7–1.8) NHL (incidence: n = 12): SIR = 0.7 (0.4–1.2)
Gustavson and Hogstedt, 1997	Sweden	Capacitor manufacture	242	1965–1991	All cancers (incidence: n = 18): SIR = 0.9 (0.5–1.4) Melanoma (incidence: n = 0): SIR = 0 (0–4.7) NHL (incidence: n = 1): SIR = 1.5 (0–8.3)
Loomis et al., 1997	USA	Electrical power	138905	1950–1988	All cancers (n = 4833): 0.9 (0.8–0.9) Malignant skin cancer (n = 116): 1.0 (0.9–1.2) NHL (n = 176): 1.0 (0.9–1.2)
Robinson et al., 1999	USA	Electrical power	30686 (M)	1982–1987	All cancers (n = 8154): PMR = 1.1 (1.0–1.1) Melanoma (n = 119): PMR = 1.2 (1.0–1.5)
Yassi et al., 2003	Canada	Transformer manufacture	2222 (M)	1969–1995	All cancers (incidence: n = 168): SIR = 1.2 (1–1.4) Melanoma (incidence: n = 10): SIR = 2.2 (1.1–4.0)
Mallin et al., 2004	Illinois, USA	Capacitor manufacture	2885	1946–2000	All cancers (n = 347): 1.1 (1.0–1.2) * Malignant skin cancer (n = 5): 1.0 (0.3–2.3)* NHL (females: n = 13): 1.8 (0.9–3.0)
Pesatori et al., 2013	Italy	Capacitor manufacture	2565	1950–2006	All cancers (n = 183): 1.0 (0.8–1.1) Melanoma (n = 1): 0.5 (0.1–3.5) NHL (n = 8): 1.7 (0.8–3.3)
Ruder et al., 2014	Indiana, USA	Capacitor manufacture	3580	1960–2008	All cancers (n = 246): 0.8 (0.7–0.9) Melanoma (n = 13): 2.6 (1.4–4.4) NHL (n = 11): 0.9 (0.5–1.7)
Ruder et al., 2014	Massachusetts, USA	Capacitor manufacture	12558	1960–2008	All cancers (n = 860): 1.0 (0.9–1.1) Melanoma (n = 8): 0.6 (0.3–1.2) NHL (n = 39): 1.2 (0.9–1.7)
Ruder et al., 2014	New York, USA	Capacitor manufacture	8727	1960–2008	All cancers (n = 801): 1.0 (0.9–1.1) Melanoma (n = 20): 1.7 (1.0–2.6) NHL (n = 19): 0.6 (0.4–0.9)

SMR = standardized mortality ratio; SIR = standardized incidence ratio; PMR = proportionate mortality ratio *computed by the authors.

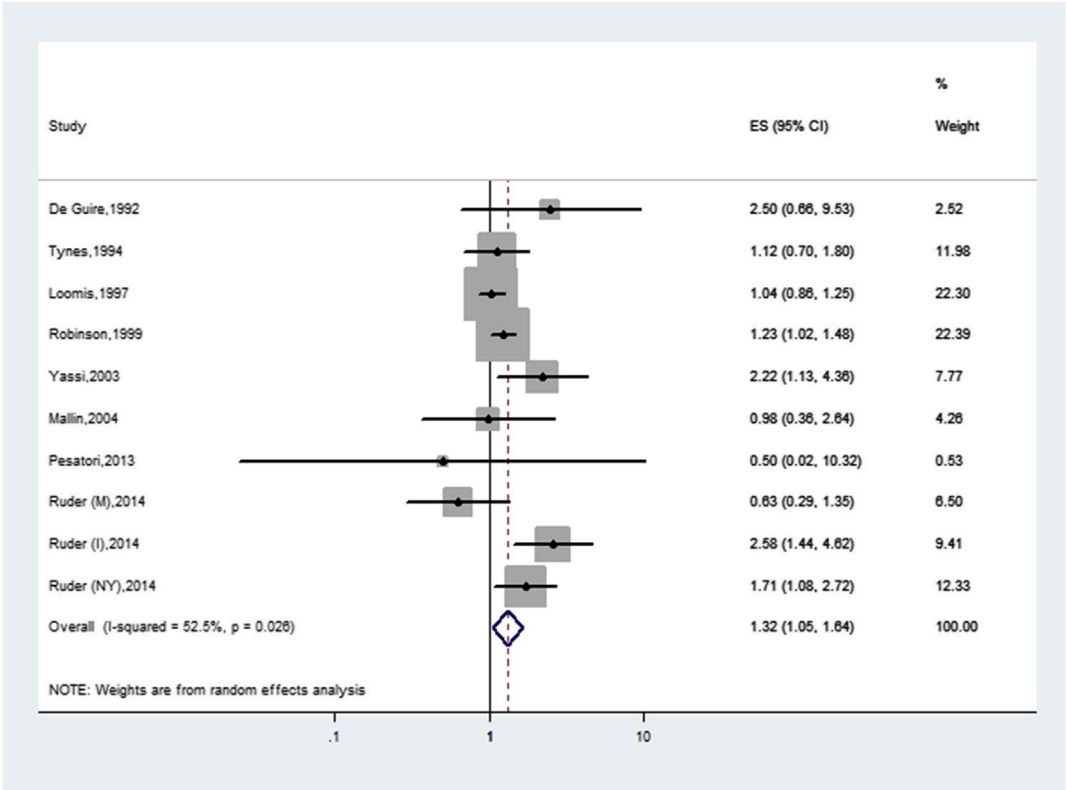

Fig. 1. Meta-analysis of cohort studies on mortality for melanoma among workers occupationally exposed to PCBs. ES: SMR estimates.

Table 2
Characteristics of the Yusho (Japan) and Yu-Cheng (Taiwan) cohort studies of subjects poisoned by contaminated rice oil, and of case-control and cohort studies carried out in the general population on cutaneous malignant melanoma and non-Hodgkin lymphoma (NHL) according to serum levels of polychlorinated biphenyls.

Authors, year	Country	Study design	Period of follow-up (years)	Type of exposure	No. subjects (cases/controls)	Outcome
Yoshimura, 2012	Japan	Cohort	1968–2007	Intoxication (rice oil)	1918	All cancers
Li et al., 2013	Taiwan	Cohort	1980–2008	Intoxication (rice oil)	1803	All cancers
Li et al., 2015	Japan and Taiwan	Cohort	1968–2008	Intoxication (rice oil)	3467	All cancers
Gallagher et al., 2011	Canada	Ca/co	2002–2008	General population	(80/310)	Cutaneous malignant melanoma
Hardell et al., 2001	Sweden	Cohort (nested Ca/Co)	1994–1997	General population	(82/83)	NHL
Quintana et al., 2004	USA	Cohort (nested Ca/Co)	1969–1983	General population	138/358	NHL
De Roos et al., 2005	USA	Ca/Co	1998–2000	General population	(97/96)	NHL
Engel et al., 2007 CLUE1	USA	Cohort (nested Ca/Co)	1970–1980	General population	74/147	NHL
Engel et al., 2007 Janus	USA	Cohort (nested Ca/Co)	1970–1980	General population	190/190	NHL
Spinelli et al., 2007	Canada	Ca/Co	2000–2004	General population	(403/459)	NHL
Cocco et al., 2008	France, Germany and Spain	Ca/Co	–	General population	(174/203)	NHL
Hardell et al., 2009	Sweden	Ca/Co	2000–2002	General population	(97/98)	NHL
Colt et al., 2009	USA	Ca/Co	1998–2000	General population	(100/100)	NHL
Laden et al., 2010	USA	Cohort (nested Ca/Co)	–	General population	145/290	NHL
Bertrand et al., 2010	USA	Cohort (nested Ca/Co)	1982–2003	General population	205/409	NHL
Viel et al., 2011	France	Ca/Co	2003–2005	General population	(34/34)	NHL
Bräuner et al., 2012	Denmark	Cohort (nested Ca/Co)	1993–1997	General population	(239/245)	NHL

Ca/Co = case-control studies.

these chemicals in the so-known "Yusho" and "Yucheng" incidents, which occurred in Japan in 1968 and in Taiwan in 1979. The estimated intakes of total PCBs and PCDFs were about 1–2 g and 3–4 g, in the episodes, respectively, with an overall daily TEQ intake in Yusho poisoning of 154 ng/kg/day, about 10000 times higher than the daily intake of the general population (1–19 pg/kg/day) (Masuda, 1996). In the Yucheng patients, the blood levels of total PCBs were up to 1156 ng/ml, about 1000 times higher than the general population (Hsu et al., 1985, 2005). Several subjects had acute effects, with dermal lesions typical of dioxin exposure, particularly chloracne, mainly due to dioxin-like PCBs and PCDFs (Hsu et al., 1985).

Two cohort studies were carried out including around 1500 Yusho and 2000 Yucheng patients, with latest updates at 43 and 30 years, respectively (Yoshimura, 2012; Li et al., 2013). No excess of skin cancer cases or deaths were reported in any cohort, so as in a meta-analysis of the two cohorts (Li et al., 2015) (Table 2).

3.3. Occupational cohort studies on exposure to PCBs and non-Hodgkin lymphoma

The occupational cohort studies included in the melanoma analysis reported also data on mortality for, or incidence of, NHL (ICD IX: 200-202) except one reporting mortality for all neoplasms of the lymphatic and hematopoietic system together (De Guire et al., 1992) and another one that did not report deaths from NHL (Robinson et al., 1999). For one of the two Canadian cohorts, we used data from a previous analysis of the cohort (Yassi et al., 1994), instead of the most recent publication (Yassi et al., 2003), which however did not report data on NHL. Therefore, we included 9 occupational cohort studies in the meta-analysis of the association of PCB exposure with NHL. None of them showed statistically significant increases of NHL risk, and the meta-analysis showed no overall association between total PCB exposure and NHL (Fig. 2), with a pooled SMR of 0.94 (95% CI: 0.73–1.23) based on a random-effects model, without significant heterogeneity (I^2 = 43.7%, p = 0.08). Based on the sensitivity analysis, the result for NHL mortality and PCBs exposure was so robust that the exclusion of any study did not change the conclusion of the meta-analysis (data not shown). There was no evidence of publication bias based on visual inspection of the funnel plot (figure not shown) in addition to Egger's test (p = 0.3).

Analysis by duration of employment was reported in 6 studies: two of them (Tynes et al., 1994; Loomis et al., 1997) showed a trend of increasing risk with increasing duration of employment or cumulative exposure, one found the lowest risk for the highest estimated cumulative PCB exposure (relative risk = 0.20) (Ruder et al., 2014) and the remaining two reported no data on NHL by duration (Mallin et al., 2004; Pesatori et al., 2013).

3.4. Population-based studies on exposure to PCBs and non-Hodgkin lymphoma

In a previous review performed by some of us (Zani et al., 2013), 13 studies (7 cohort and 6 case-control studies) which evaluated PCB concentration in blood or abdominal subcutaneous fat were selected (Hardell et al., 2001; Quintana et al., 2004; De Roos et al., 2005; Engel et al., 2007; two cohorts; Spinelli et al., 2007; Cocco et al., 2008; Colt et al., 2009; Hardell et al., 2009; Bertrand et al., 2010; Laden et al., 2010; Viel et al., 2011; Bräuner et al., 2012). One paper reported the findings from three separate cohorts (Engel et al., 2007), but only two of them were included in the analysis (CLUE1 and Janus studies), because an update of the data was reported for the third one separately (Laden et al., 2010).

An update of our search for articles published since 2013 and a comparison with the IARC Monograph provided one further study on the risk of B-cell NHL by PCB exposure, consisting in a case-control study nested within two prospective cohorts (Kelly et al., 2017). During 16 years follow-up, 270 incident cases and 270 healthy controls were recruited in the study and their pre-diagnostic plasma levels of various organochlorines, including 6 PCB congeners were measured (Kelly et al., 2017). This study, however, could not be added to the others included in the meta-analysis because only separate analyses for each PCB congener are reported. The study found no evidence of an association of plasma levels of each PCB congener with B-cell NHL, with a paradoxical inverse trend of decreasing risk for increasing PCB plasma levels for some congeners.

All the cohort studies included in the meta-analysis used a nested case-control or case-cohort design (Table 2) and provided

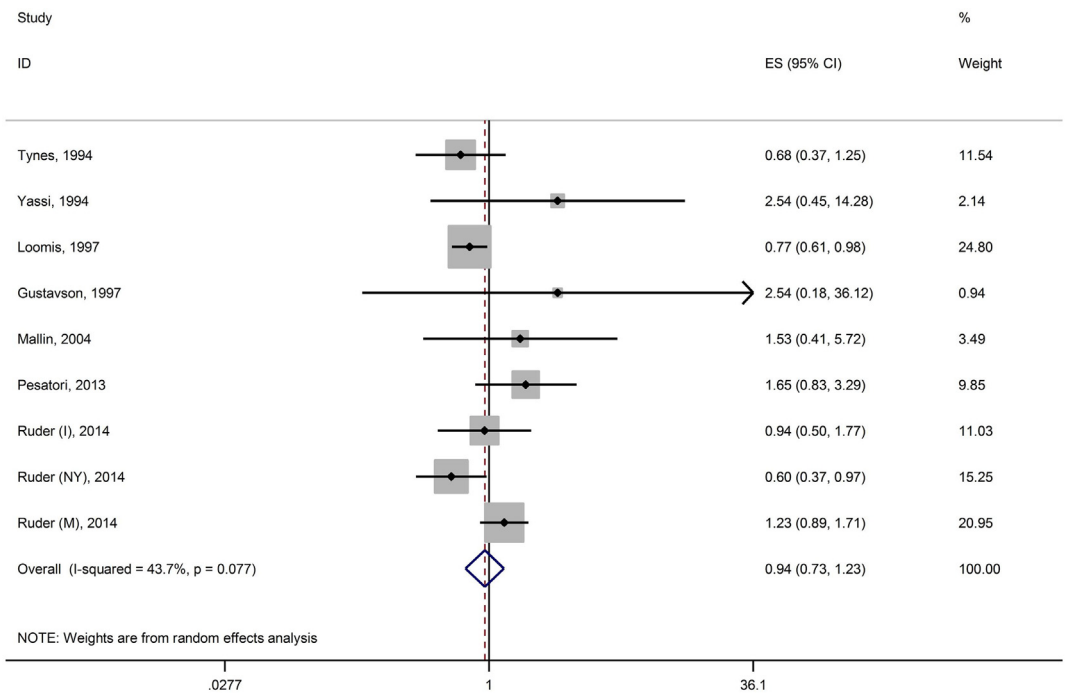

Fig. 2. Meta-analysis of cohort studies on mortality for non-Hodgkin lymphoma among workers occupationally exposed to PCBs. ES: SMR estimates.

ORs as measures of association. Most studies provided estimates controlled for possible confounders including demographic and socio-economic factors, alcohol drinking, tobacco smoking, BMI, family history of NHL and others. Some studies also provided OR estimates for the main PCB congeners, and for subgroups of congeners according to functional groups (Cocco et al., 2008) or level of chlorination (Hardell et al., 2009).

Two of the 7 cohort studies and four of the 6 case-control studies reported a statistically significant trend of increasing risk of NHL with increasing PCB concentration, or an increased risk for levels higher than, compared to lower than, the median of PCB distribution (Engel et al., 2007; Spinelli et al., 2007; Colt et al., 2009; Hardell et al., 2009; Bertrand et al., 2010; Viel et al., 2011). A meta-analysis of these studies was performed comparing the highest vs lowest quantile or higher vs lower than the median of PCB distribution, based on all the studies apart from those by Viel et al. (2011) and by Colt et al. (2009) because they reported PCB serum level only as a continuous variable. A summary OR = 1.5 (95% CI: 1.1–1.7) was found, with no evidence of heterogeneity ($I^2 = 22.5\%$, $p > 0.1$) and publication bias, and similar results in both cohort and case-control studies (Zani et al., 2013). A meta-analysis of the immunotoxic PCB congeners 138, 153 and 180, based on 6-8 cohort and case-control studies reporting suitable data, provided results consistent with the main analysis (Zani et al., 2013).

Furthermore, we analyzed the dose-response relationship between total PCB serum levels and OR of NHL on the basis of the ORs reported for each quantile of PCB serum levels, using the median values for each quantile for studies showing ORs for at least three categories of PCB exposure. Seven of the 13 studies were excluded from this analysis: three reported PCB concentrations in subcutaneous adipose tissue, which were higher than blood levels (Hardell et al., 2001; Quintana et al., 2004; Bräuner et al., 2012), two provided ORs for only two categories of PCB levels (above and below the median) (Hardell et al., 2009; Viel et al., 2011), two did not provide ORs for total PCBs (De Roos et al., 2005; Colt et al., 2009). The findings from the remaining six studies (Engel et al., 2007; two cohorts; Spinelli et al., 2007; Cocco et al., 2008; Bertrand et al., 2010; Laden et al., 2010) are shown in Fig. 3. Apart from one study with extremely high ORs for relatively low PCB values (Engel et al., 2007; CLUE1 study), the others showed no OR increase until PCB serum levels of 1000 ng/g lipid. A meta-analysis of these studies provided an overall OR = 1.42 (95% CI: 1.10–1.83) for PCB serum levels around 1000 ng/g lipid, compared to values lower than, or next to, 500 ng/g lipid, with low heterogeneity among the studies ($I^2 = 9.6\%$, $p = 0.4$). Too few data were available for calculating ORs for PCB serum levels above 1000 ng/g lipid.

The Yucheng and Yusho mortality studies on patients accidentally exposed to high doses of PCBs and PCDFs showed contrasting results (Yoshimura, 2012; Li et al., 2013, 2015). Among Yucheng patients, the SMR for cancer of lymphatic and haematopoietic tissue was increased in males (SMR = 3.0, based on 5 deaths observed) but not in females (no death observed vs 1.6 expected); in the Yusho cohort no excess in haematopoietic malignancies was found. However, the neoplasms of lymphatic and haematopoietic tissue in the Yucheng cohort included also leukemias, and no separate data for NHL were available (Yoshimura, 2012; Li et al., 2013). A meta-analysis of both cohorts did not report an increased risk for neoplasms of the lymphatic and haematopoietic tissue (Li et al., 2015).

4. Discussion

Our systematic review and meta-analysis of the association between PCB exposure and melanoma and NHL showed contrasting results when considering a) cohort studies on subjects professionally exposed to PCBs, b) cohort and case-control studies on unexposed people with individual measures of exposure, and c) cohort studies on people undergoing mass poisoning by PCBs and PCDFs.

4.1. Cutaneous melanoma

The meta-analysis of the occupational cohort studies showed a summary SMR of 1.32, suggesting a moderate increase of the risk

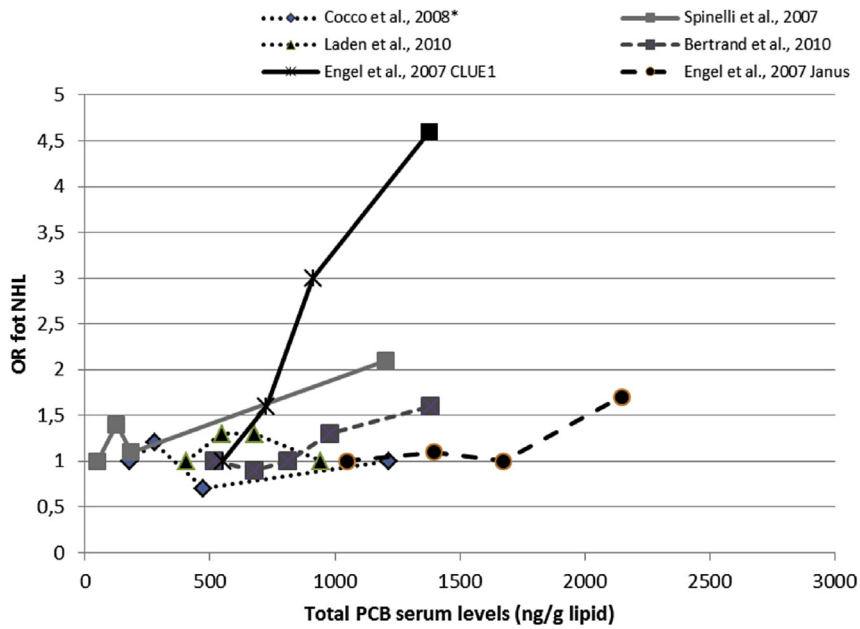

Fig. 3. Odds ratios (ORs) of non-Hodgkin lymphoma (NHL) according to total PCB serum levels in cohort and case-control studies proving data for at least three categories of PCB levels.

for occupational exposure. However, the dose-response analysis provided inconsistent results, with only two studies showing an increase of the risk of melanoma with increasing duration of employment or cumulative exposure. It is noteworthy that the largest cohort study carried out among capacitor manufacture workers, including a pooled analysis of three cohorts, some of which had been highly exposed to PCBs, showed the lowest risk for the highest level of exposure (Ruder et al., 2014), estimated by validated semi-quantitative job-exposure matrices (Hopf et al., 2014a, 2014b). Furthermore, a recent meta-analysis on occupational exposure to PCBs and risk of melanoma does not support the hypothesis of an association between exposure to PCBs and incidence of, or mortality from, malignant melanoma (Boffetta et al., 2016).

One only case-control study on cutaneous melanoma and PCB exposure using PCB serum levels as individual measures of exposure was carried out so far (Gallagher et al., 2011). The study by Gallagher et al. (2011) showed an increased risk of cutaneous melanoma for PCB exposure, with a dose-response trend. However, it is relatively small (80 cases) and although it took account for sun exposure and other potential confounders, suffers from some drawbacks: the cases and controls came from two previous, different studies, with only an approximate age and sex matching; the retrospective design of the research, with PCB serum levels measured in melanoma cases after the beginning of the disease; and finally, more importantly, the fact that PCB analysis was performed on frozen serum samples some years after having collected them, with longer interval time between collection and laboratory assay for cases than controls (4-8 and 1–5 years, respectively). In fact, storage of serum in plastic containers has been found to increase total PCB concentrations (Karmaus and Riebow, 2004), raising suspicion that higher PCB levels in sera of cases than controls may be due, partly, to longer interval between collection and laboratory analysis.

Both Yusho and Yucheng cohort studies on people intoxicated by rice oil containing PCBs and PCDFs found no excess of deaths for skin cancer (Yoshimura, 2012; Li et al., 2013, 2015). Likewise, no excess of skin melanoma was observed in the population living in Michalovce, eastern Slovakia, an area highly polluted by a PCB-producing factory, with geometric means of total PCB serum levels of 3327.6 and 2751.8 ng/g lipid, in male and female adults, respectively, almost three-fold higher than a control population living in an unpolluted area of the same region (Pavuk et al., 2004).

In a case–control study from nine European countries, Behrens et al. (2010) investigated the association between uveal melanoma and exposure to PCBs, which was assigned if subjects reported occupational exposure to transformer/capacitor oils at personal interviews. Analyses were adjusted for age, country, eye color, and history of ocular damage from ultraviolet light, though no information was available on lifetime sun exposure. Only men reported exposure to transformer/capacitor oils. In men, the OR for any exposure was 2.74 (99.3% CI, 1.07–7.02). However, a re-analysis of the same study using job-specific questionnaires to solicit detailed questions on occupational tasks, which have higher specificity on the assessment of job-related exposures as compared to the other tools, found no statistically significant increased risk of uveal melanoma for exposure to PCBs (OR = 1.23, 95% CI: 0.55–2.74) (Behrens et al., 2012).

A recent review (Fortes and De Vries, 2008) suggested a role for some specific work-related exposures in aetiology of cutaneous melanoma. However, they included in the analysis also studies in which the subjects had been exposed to different chemicals (solvent, PAH, pesticides, etc.), thus not allowing to evaluate the association between exposure to PCBs and risk for melanoma properly.

4.2. Non-Hodgkin lymphoma

None of occupational cohort studies showed statistically significant excess of NHL or all lymphomas together, with a summary SMR = 0.94. Accordingly, analysis by duration of employment or cumulative exposure to PCBs provided no evidence of association. Among the 11 cohort and case-controls studies with individual PCB measures, only one showed a statistically significant increased risk of NHL. However, a meta-analysis of these studies provided an overall statistically significant OR = 1.40 (Zani et al., 2013). The

dose-response analysis based on six studies showed an increased risk of NHL for PCB serum levels around 1000 ng/g lipid, compared to lower than or next to 500 ng/g lipid, with an overall OR = 1.42, with low heterogeneity among the studies. No data were available for evaluation of NHL for higher PCB serum levels. Although reverse causality may theoretically bias the results of case-control studies toward positive findings, as serum samples were collected after the disease occurrence, this is unlikely to have occurred in most case-control studies on melanoma or NHL and PCB, because incidence cases were enrolled before starting chemotherapy. Anyway, the meta-analytic results of cohort studies, in which sera collection precedes the disease occurrence, and case-control studies are very similar (Zani et al., 2013).

Freeman and Kohles (2012) recently conducted a meta-analysis of six case-control studies on NHL risk according to serum levels of 10 PCB congeners, and computed a summary statistically significant OR = 1.43 for the sum of these congeners (95% CI: 1.31–1.55). Another review (Kramer et al., 2012) concluded that there was sufficient evidence supporting the causal role of PCBs in the genesis of lymphomas, the most important results coming from population studies. The commonly hypothesized mechanism for development of NHL due to PCB exposure is immune dysregulation able to decrease the resistance of the subject, the control of cell proliferation and differentiation. Indeed, both toxicological and epidemiological data showed immunotoxic effects of PCBs, particularly immunosuppression, which has been recognized as the most important risk factor for NHL development (Kramer et al., 2012; Strauss and Heiger-Bernays, 2012). However, the Yusho and Yucheng cohort studies showed inconsistent results, with an excess of neoplasms of lymphatic and hematopoietic tissue, including some leukemia cases, only in males in one cohort (Li et al., 2013, 2015; Yoshimura, 2012). Accordingly, no increased incidence of NHL was found in a population living in a highly PCB polluted area in Slovakia, with around three-fold elevated PCB serum levels in adults compared to unexposed reference population (Pavuk et al., 2004). Finally, a recent case-control study nested within two cohort studies, with measurements of 6 PCB congeners carried out on pre-diagnostic plasma levels, provided negative results for all congeners, showing no evidence that a higher body burden of PCBs increased the risk of subsequent NHL diagnosis (Kelly et al., 2017).

Furthermore, the positive results of some case-control studies on melanoma and NHL based on PCB serum levels may be due to reserve causality, because the cases' serum samples were collected after the occurrence of the disease. Particularly, weight loss due to cancer and/or chemotherapy may lead to increases in PCB serum levels as chemicals accumulated in body fat are mobilized. However, incidence cases were usually collected in these studies before they underwent chemotherapy in most of them.

4.3. A comparison of this review with the IARC Working Group evaluation and the Boffetta et al. meta-analysis

Our systematic review and meta-analysis of the risk of melanoma and NHL for PCB exposure provided inconsistent results between occupational cohort studies, population-based studies with individual measures and studies on highly exposed populations, not allowing to draw firm conclusions on these associations. These conclusions are somewhat different from both the IARC Working Group evaluation and the systematic review on PCB exposure and malignant melanoma by Boffetta et al. (2016).

The IARC Working Group performed a comprehensive review of the overall evidence for carcinogenicity of PCBs, on the basis of human carcinogenicity data, animal carcinogenicity and mechanistic data (IARC, 2016; Lauby-Secretan et al., 2016). The Working Group concluded for sufficient evidence for carcinogenicity of PCBs in humans for malignant melanoma and limited evidence for NHL and breast cancer (Lauby-Secretan et al., 2013; IARC, 2016).

However, some methodological issues of the Working Group evaluation concerning selection and assessment of epidemiologic studies should be considered.

Particularly, the data on malignant melanoma and PCB exposure provided a substantial support for the Working Group conclusion of sufficient evidence for PCB carcinogenicity in humans. They were mainly based on the results of 7 occupational cohort studies, 6 of which were positive, as reported in the Summary of the Monograph (IARC, 2016). However, the criteria used for retrieving and selecting these 7 occupational studies were undefined. On the other hand, we retrieved 11 occupational cohort studies, only four of which showed a statistically significant increased risk of skin cancer. Further support comes from the evaluation of exposure-response relationship: the study reporting a statistically significant exposure-response linear trend in electric power workers was mentioned in the Summary (Loomis et al., 1997). However, the joined and updated analysis of the three NIOSH cohorts of workers at electrical capacitor manufacturing plants showing the lowest risk of melanoma for the highest level of estimated exposure (Ruder et al., 2014) was ignored.

The Working Group based its PCB evaluation also on the case-control study on skin melanoma by Gallagher et al. (2011) and on the case-control study on uveal melanoma by Behrens et al. (2010), which showed a positive association with the history of occupational exposure to PCBs. However, a subsequent analysis of the Behrens et al. study using a more specific tool for defining PCB occupational exposure did not show a significantly increased risk; these findings were not considered in the final evaluation of the Working Group (Behrens et al., 2012). Also the findings from the two Asian cohorts of poisoned subjects with an overall intake of PCBs 100-10000 times higher than the general population, that did not report an increased risk for skin cancer and showed inconsistent results for NHL (Yoshimura, 2012; Li et al., 2013, 2015) were ignored in the Summary.

Finally, it is noteworthy that the conclusions of the IARC Working Group on NHL took into account “study size and quality”, although the criteria for evaluating study size and quality were undefined.

The study of Boffetta et al. (2016) aimed to carry out a meta-analysis of studies on exposure to PCBs and the risk of malignant melanoma. They included both occupational cohort studies and studies carried out in “populations living in areas with reported PCB exposure”. The occupational cohort studies included in Boffetta et al. (2016) and our meta-analyses are substantially the same, although Boffetta et al. used the combined analysis of the three cohorts of capacitor-manufacturing workers exposed to PCBs at plants in Indiana, Massachusetts, and New York (Ruder et al., 2014), whereas we considered the three cohorts separately. Among the population-based studies, Boffetta et al. included an incidence study on the health effects of exposure to PCBs and metals in a contaminated community in Sweden (Helmfrid et al., 2012), a study on residents in an area in eastern Slovakia with extensive environmental contamination from a PCB production site (Michalovce) (Pavuk et al., 2004) and a mortality and cancer incidence study in cohorts of Swedish fishermen and fishermen's wives (Mikoczy and Rylander, 2009). These three studies, however, did not provide data on individual PCB exposure, and therefore were not included in our review. Because of the lack of population-based studies on the association between malignant melanoma and PCB exposure with individual measures of exposure, apart from the Gallagher et al. study (2011), we could not perform a meta-analysis of non-occupational studies, therefore our findings are not properly comparable with those in Boffetta

et al. (2016).

In conclusion, these findings do not provide a strong evidence that PCB exposure can increase the risk of melanoma and NHL in humans.

Disclosure statement

The authors report no conflicts of interest. The author's affiliation is as shown on the cover page. The authors have sole responsibility for the writing and content of the paper. No funding was available for this study.

Acknowledgements

We are grateful to internal postgraduate medical doctors attending the School of Hygiene and Preventive Medicine of the University of Brescia, Dr. Francesca Bianchi, Dr. Marco Moneda and Dr. Giuseppe Zenoni, who collaborated in searching and collecting the studies of interest.

Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.chemosphere.2017.05.053.

References

ATSDR, 2003. Health Consultation Concerning Polychlorinated Biphenyls, Dioxins, and Pesticides in Soil, Blood and Air from Anniston. Agency for Toxic Substances and Disease Registry, Alabama.
Behrens, T., Lynge, E., Cree, I., Lutz, J.M., Eriksson, M., Guénel, P., Merletti, F., Morales-Suarez-Varela, M., Afonso, N., Stengrevics, A., Stang, A., et al., 2012. Occupational exposure to endocrine-disrupting chemicals and the risk of uveal melanoma. Scand. J. Work Environ. Health 38, 476–483.
Behrens, T., Lynge, E., Cree, I., Sabroe, S., Lutz, J.M., Afonso, N., Eriksson, M., Guénel, P., Merletti, F., Morales-Suarez-Varela, M., Stengrevics, A., et al., 2010. Occupational exposure to electromagnetic fields and sex-differential risk of uveal melanoma. Occup. Environ. Med. 67, 751–759.
Bertrand, K.A., Spiegelman, D., Aster, J.C., Altshul, L.M., Korrick, S.A., Rodig, S.J., Zhang, S.M., Kurth, T., Laden, F., 2010. Plasma organochlorine levels and risk of non-Hodgkin lymphoma in a cohort of men. Epidemiology 21, 172–180.
Boffetta, P., Catalani, S., Tomasi, C., Pira, E., Apostoli, P., 2016. Occupational exposure to polychlorinated biphenyls and risk of melanoma: a meta-analysis. Eur. J. Cancer Prev. http://dx.doi.org/10.1097/CEJ.0000000000000316.
Bradburn, M.J., 2004. Updated and New Commands for Meta-analysis in STATA. Cancer Research UK Medical Statistics Group, Centre for Statistics in Medicine, Oxford.
Bräuner, E.V., Sørensen, M., Gaudreau, E., LeBlanc, A., Eriksen, K.T., Tjønneland, A., Overvad, K., Raaschou-Nielsen, O., 2012. A prospective study of organochlorines in adipose tissue and risk of non-Hodgkin lymphoma. Environ. Health Perspect. 120, 105–111.
Breslow, N.E., Day, N.E., 1987. Statistical Methods in Cancer Research. Volume 11-The Design and Analysis of Cohort Studies. International Agency for Research on Cancer (IARC) Sci Publ, Lyon.
Cocco, P., Brennan, P., Ibba, A., de Sanjosé Llongueras, S., Maynadié, M., Nieters, A., Becker, N., Ennas, M.G., Tocco, M.G., Boffetta, P., 2008. Plasma polychlorobiphenyl and organochlorine pesticide level and risk of major lymphoma subtypes. Occup. Environ. Med. 65, 132–140.
Colt, J.S., Rothman, N., Severson, R.K., Hartge, P., Cerhan, J.R., Chatterjee, N., Cozen, W., Morton, L.M., De Roos, A.J., Davis, S., et al., 2009. Organochlorine exposure, immune gene variation, and risk of non-Hodgkin lymphoma. Blood 113, 1899–1905.
De Guire, L., Cyr, D., Thériault, G., Provencher, S., Iturra, H., Case, B.W., 1992. Malignant melanoma of the skin among workers in a telecommunications industry: mortality study 1976-83. Br. J. Ind. Med. 49, 728–731.
De Roos, A.J., Hartge, P., Lubin, J.H., Colt, J.S., Davis, S., Cerhan, J.R., Severson, R.K., Cozen, W., Patterson Jr., D.G., Needham, L.L., Rothman, N., 2005. Persistent organochlorine chemicals in plasma and risk of non-Hodgkin's lymphoma. Cancer Res. 65, 11214–11226.
Dersimonian, R., Laird, N., 1986. Meta-analysis in clinical trials. Control Clin. Trials 7, 177–188.
Egger, M., Schneider, M., Davey Smith, G., 1998. Spurious precision? Meta-analysis of observational studies. BMJ 316, 140–144.
Engel, L.S., Laden, F., Andersen, A., Strickland, P.T., Blair, A., Needham, L.L., Barr, D.B., Wolff, M.S., Helzlsouer, K., Hunter, D.J., et al., 2007. Polychlorinated biphenyl levels in peripheral blood and non-Hodgkin's lymphoma: a report from three cohorts. Cancer Res. 67, 5545–5552.
Fortes, C., De Vries, E., 2008. Nonsolar occupational risk factors for cutaneous melanoma. Int. J. Dermatol 47, 319–328.
Freeman, M.D., Kohles, S.S., 2012. Plasma levels of polychlorinated biphenyls, non-Hodgkin lymphoma, and causation. J. Environ. Public Health 2012, 258981.
Gallagher, R.P., Macarthur, A.C., Lee, T.K., Weber, J.P., Leblanc, A., Mark Elwood, J., Borugian, M., Abanto, Z., Spinelli, J.J., 2011. Plasma levels of polychlorinated biphenyls and risk of cutaneous malignant melanoma: a preliminary study. Int. J. Cancer 128, 1872–1880.
Golden, R., Kimbrough, R., 2009. Weight of evidence evaluation of potential human cancer risks from exposure to polychlorinated biphenyls: an update based on studies published since 2003. Crit. Rev. Toxicol. 39, 299–331.
Gustavsson, P., Hogstedt, C., 1997. A cohort study of Swedish capacitor manufacturing workers exposed to polychlorinated biphenyls (PCBs). Am. J. Ind. Med. 32, 234–239.
Hardell, E., Eriksson, M., Lindström, G., Van Bavel, B., Linde, A., Carlberg, M., Liljegren, G., 2001. Case-control study on concentrations of organohalogen compounds and titers of antibodies to Epstein-Barr virus antigens in the etiology of non-Hodgkin lymphoma. Leuk. Lymphoma 42, 619–629.
Hardell, K., Carlberg, M., Hardell, L., Björnfoth, H., Ericson Jogsten, I., Eriksson, M., Van Bavel, B., Lindström, G., 2009. Concentrations of organohalogen compounds and titres of antibodies to Epstein-Barr virus antigens and the risk for non-Hodgkin lymphoma. Oncol. Rep. 21, 1567–1576.
Higgins, J.P., Thompson, S.G., 2002. Quantifying heterogeneity in a meta-analysis. Stat. Med. 21, 1539–1558.
Helmfrid, I., Berglund, M., Löfman, O., Wingren, G., 2012. Health effects and exposure to polychlorinated biphenyls (PCBs) and metals in a contaminated community. Environ. Int. 44, 53–58.
Hopf, N.B., Ruder, A.M., Succop, P., Waters, M.A., 2014a. Evaluation of cumulative PCB exposure estimated by a job exposure matrix versus PCB serum concentrations. Environ. Sci. Pollut. Res. Int. 21, 6314–6323.
Hopf, N.B., Ruder, A.M., Waters, M.A., 2014b. Historical reconstruction of polychlorinated biphenyl (PCB) exposures for workers in a capacitor manufacturing plant. Environ. Sci. Pollut. Res. Int. 21, 6419–6433.
Hsu, S.T., Ma, C.I., Hsu, S.K., Wu, S.S., Hsu, N.H., Yeh, C.C., Wu, S.B., 1985. Discovery and epidemiology of PCB poisoning in Taiwan: a four-year followup. Environ. Health Perspect. 59, 5–10.
Hsu, J.F., Guo, Y.L., Yang, S.Y., Liao, P.C., 2005. Congener profiles of PCBs and PCDD/Fs in Yucheng victims fifteen years after exposure to toxic rice-bran oils and their implications for epidemiologic studies. Chemosphere 61, 1231–1243.
IARC, 2016. Polychlorinated biphenyls. In: IARC Monographs on the Evaluation of Carcinogenic Risks to Humans, vol. 107. International Agency for Research on Cancer (Polychlorinated and Polybrominated Biphenyls. Lyon, France).
Kelly, R.S., Kiviranta, H., Bergdahl, I.A., Palli, D., Johansson, A.S., Botsivali, M., Vineis, P., Vermeulen, R., Kyrtopoulos, S.A., Chadeau-Hyam, M., EnvirOGenoMarkers project consortium, 2017. Prediagnostic plasma concentrations of organochlorines and risk of B-cell non-Hodgkin lymphoma in envirogenomarkers: a nested case-control study. Environ. Health 16, 9.
Karmaus, W., Riebow, J.F., 2004. Storage of serum in plastic and glass containers may alter the serum concentration of polychlorinated biphenyls. Environ. Health Perspect. 112, 643–647.
Kramer, S., Hikel, S.M., Adams, K., Hinds, D., Moon, K., 2012. Current status of the epidemiologic evidence linking polychlorinated biphenyls and non-Hodgkin lymphoma, and the role of immune dysregulation. Environ. Health Perspect. 120, 1067–1075.
Laden, F., Bertrand, K.A., Altshul, L., Aster, J.C., Korrick, S.A., Sagiv, S.K., 2010. Plasma organochlorine levels and risk of non-Hodgkin lymphoma in the Nurses' Health Study. Cancer Epidemiol. Biomarkers Prev. 19, 1381–1384.
Lauby-Secretan, B., Loomis, D., Grosse, Y., El Ghissassi, F., Bouvard, V., Benbrahim-Tallaa, L., Guha, N., Baan, R., Mattock, H., Straif, K., WHO International Agency for Research on Cancer, 2013. Carcinogenicity of polychlorinated biphenyls and polybrominated biphenyls. Lancet Oncol. 14, 287–288.
Lauby-Secretan, B., Loomis, D., Baan, R., El Ghissassi, F., Bouvard, V., Benbrahim-Tallaa, L., Guha, N., Grosse, Y., Straif, K., 2016. Use of mechanistic data in the IARC evaluations of the carcinogenicity of polychlorinated biphenyls and related compounds. Environ. Sci. Pollut. Res. Int. 23, 2220–2229.
Li, M.C., Tsai, P.C., Chen, P.C., Hsieh, C.J., Leon Guo, Y.L., Rogan, W.J., 2013. Mortality after exposure to polychlorinated biphenyls and dibenzofurans: 30 years after the "Yucheng accident". Environ. Res. 120, 71–75.
Li, M.C., Chen, P.C., Tsai, P.C., Furue, M., Onozuka, D., Hagihara, A., Uchi, H., Yoshimura, T., Guo, Y.L., 2015. Mortality after exposure to polychlorinated biphenyls and polychlorinated dibenzofurans: a meta-analysis of two highly exposed cohorts. Int. J. Cancer 137, 1427–1432.
Loomis, D., Browning, S.R., Schenck, A.P., Gregory, E., Savitz, D.A., 1997. Cancer mortality among electric utility workers exposed to polychlorinated biphenyls. Occup. Environ. Med. 54, 720–728.
Mallin, K., McCann, K., D'Aloisio, A., Freels, S., Piorkowski, J., Dimos, J., Persky, V., 2004. Cohort mortality study of capacitor manufacturing workers, 1944-2000. J. Occup. Environ. Med. 46, 565–576.
Masuda, Y., 1996. Approach to risk assessment of chlorinated dioxins from Yusho PCB poisoning. Chemosphere 32, 583–594.
Mikoczy, Z., Rylander, L., 2009. Mortality and cancer incidence in cohorts of Swedish fishermen and fishermen's wives: updated findings. Chemosphere 74, 938–943.
Pavuk, M., Cerhan, J.R., Lynch, C.F., Schecter, A., Petrik, J., Chovancova, J., Kocan, A.,

2004. Environmental exposure to PCBs and cancer incidence in eastern Slovakia. Chemosphere 54, 1509–1520.
Pesatori, A.C., Grillo, P., Consonni, D., Caironi, M., Sampietro, G., Olivari, L., Ghisleni, S., Bertazzi, P.A., 2013. Update of the mortality study of workers exposed to polychlorinated biphenyls (PCBs) in two Italian capacitor manufacturing plants. Med. Lav. 104, 107–114.
Quintana, P.J., Delfino, R.J., Korrick, S., Ziogas, A., Kutz, F.W., Jones, E.L., Laden, F., Garshick, E., 2004. Adipose tissue levels of organochlorine pesticides and polychlorinated biphenyls and risk of non-Hodgkin's lymphoma. Environ. Health Perspect. 112, 854–861.
Robinson, C.F., Petersen, M., Palu, S., 1999. Mortality patterns among electrical workers employed in the U.S. construction industry, 1982-1987. Am. J. Ind. Med. 36, 630–637.
Ruder, A.M., Hein, M.J., Hopf, N.B., Waters, M.A., 2014. Mortality among 24,865 workers exposed to polychlorinated biphenyls (PCBs) in three electrical capacitor manufacturing plants: a ten-year update. Int. J. Hyg. Environ. Health 217, 176–187.
Spinelli, J.J., Ng, C.H., Weber, J.P., Connors, J.M., Gascoyne, R.D., Lai, A.S., Brooks-Wilson, A.R., Le, N.D., Berry, B.R., Gallagher, R.P., 2007. Organochlorines and risk of non-Hodgkin lymphoma. Int. J. Cancer 121, 2767–2775.
Strauss, H.S., Heiger-Bernays, W., 2012. Methodological limitations may prevent the observation of non-Hodgkin's lymphoma in bioassays of polychlorinated biphenyls. Toxicol. Pathol. 40, 995–1003.
Tobias, A., 1999. Assessing the influence of a single study in meta-analysis. Stata Tech. Bull. 47, 15–17.
Tynes, T., Reitan, J.B., Andersen, A., 1994. Incidence of cancer among workers in Norwegian hydroelectric power companies. Scand. J. Work Environ. Health 20, 339–344.
Viel, J.F., Floret, N., Deconinck, E., Focant, J.F., De Pauw, E., Cahn, J.Y., 2011. Increased risk of non-Hodgkin lymphoma and serum organochlorine concentrations among neighbors of a municipal solid waste incinerator. Environ. Int. 37, 449–453.
Wong, O., Raabe, G.K., 1996. Application of meta-analysis in reviewing occupational cohort studies. Occup. Environ. Med. 53, 793–800.
Yassi, A., Tate, R., Fish, D., 1994. Cancer mortality in workers employed at a transformer manufacturing plant. Am. J. Ind. Med. 25, 425–437.
Yassi, A., Tate, R.B., Routledge, M., 2003. Cancer incidence and mortality in workers employed at a transformer manufacturing plant: update to a cohort study. Am. J. Ind. Med. 44, 58–62.
Yoshimura, T., 2012. Yusho: 43 years later. Kaohsiung J. Med. Sci. 28, S49–S52.
Zani, C., Toninelli, G., Filisetti, B., Donato, F., 2013. Polychlorinated biphenyls and cancer: an epidemiological assessment. J. Environ. Sci. Health C Environ. Carcinog. Ecotoxicol. Rev. 31, 99–144.